CHAPTER 7

Topos and Sheaf Theory

"L'état de vérité partiel serait alors l'état de silence intérieur et d'écoute."
– Alexander Grothendieck in La Clef des Songes

"I do not approve. I understand." – Mr. Spock to Anan 7

7.1. In the previous chapter, our intention was to introduce a few objects and mathematical notions like that of gluing (for example that of projective space from affine charts), local rings, and coordinate rings. In this chapter, our goal is to introduce the notion of topological spaces, shaves, ringed spaces and topoi which make up interesting part of EGA and SGA.

For reasons, we will bi-fold this chapter into Part I which will be about the local-to-global language of algebraic geometry and Part II which will the categorical viewpoint towards the notion of space.

Part I: The local-to-global language

7.2. The best point to start this local-to-global language is the *sheaf theory* which was precisely developed by Leray[1] [**32**] for analyzing the obstruction of piecing together local data to get global data.

1. Presheaf and Sheaf

Definition 7.3. Let X be a topological space. We define presheaf $\mathcal{F}$ on X with the assignment of $\mathcal{F}(U)$ to each open set $U \subset X$ such that for every inclusion of opens $V \subset U$, we have restriction map

$$\text{res}_U^V : \mathcal{F}(U) \to \mathcal{F}(V) \tag{195}$$

such that

This is a part of Prelude of Schemes project available at https://aayushayh.github.io/PtoS.html.
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[1]It is also interesting to note that Leray developed this 1946 while he was a prisoner prisoner of war in Oflag XVII-A in Austria during World War II.

(1) $\text{res}^U_U = id_{\mathcal{F}(U)}$ for all the opens U
(2) Whenever $W \subset V \subset U$, we have

$$\text{res}^W_V \cdot \text{res}^V_U = \text{res}^W_U \tag{196}$$

which is equivalent to saying that the following commute

$$\begin{array}{ccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}(W) \\ & \searrow \quad \nearrow & \\ & \mathcal{F}(V) & \end{array}$$

Definition 7.4 (Morphism of presheaves). A morphism of presheaves $\mathcal{F}, \mathcal{G}$ on X is a collection of maps $\alpha : \mathcal{F}(U) \to \mathcal{G}(U)$ for each open open set on X such that whenever there is inclusion of opens sets, the following commutes

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\alpha(U)} & \mathcal{G}(U) \\ {\scriptstyle res^V_U}\downarrow & & \downarrow{\scriptstyle res^V_U} \\ \mathcal{F}(V) & \xrightarrow[\alpha(V)]{} & \mathcal{G}(V) \end{array}$$

whenever $V \subset U$.

7.5. In many places, one defines an element f in $\mathcal{F}(U)$ to be the *section* over U. We can further simply $\text{res}^V_U(f)$ as $\sigma|_V$.

Definition 7.6 (Presheaf as a functor). In the language of categories, we can define a presheaf (of sets) on $\mathcal{C}$ to be a 'contravariant' functor from the category $\mathcal{C}$ (in our previous examples, $\mathcal{C}$ is the category of open subsets of a fixed space) to the category of sets (again, it can be any other category).

The category of presheaves (of sets) is denoted by $PSh(\mathcal{C})_{Sets}$. The morphisms in this category are the natural transformations mentioned in Def. 7.4.

7.7 (EGA1 3.1.5). If we have a morphism of sheaf $\alpha : \mathcal{F} \to \mathcal{G}$ on X, then for $U \subset X$, there is a morphism defined on the level of $\alpha_U : \mathcal{F}|_U \to \mathcal{G}|_U$.

In particular, if we have two sheaves $\mathcal{F}, \mathcal{G}$ on X, then we can define the 'sheaf $\mathcal{H}om$' the sheaf

$$\mathcal{H}om(\mathcal{F}, \mathcal{G})(U) = \text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U) \tag{197}$$

where $\text{Hom}(\mathcal{F}, \mathcal{G})$ is the set of morphism.

Remark 7.8. Since $\mathcal{F}$ is a contravariant functor, the morphism of sheaves is a natural transformation.

Note that the category of the presheaves on $\mathcal{C}$ the functor category[2], denoted as $PSh(\mathcal{C})$. The category of sheaves $Sh(\mathcal{C})$ on $\mathcal{C}$ is the full subcategory of $PSh(\mathcal{C})$.

Definition 7.9 (Sheaf). A presheaf $\mathcal{F}$ on X is sheaf if it satisfies the following axioms

(1) For any open $U \subset X$ and any open covering $\{U_i\}_{i\in I}$ of U and any sections $f_1, f_2 \in \mathcal{F}(U)$ such that $f_1|u_i = f_2|u_i$ for all i, then $f_1 = f_2$.
    This axiom is called *identity axiom*. A presheaf which satisfies this axiom is called a *separated sheaf* (or sometimes, a 'decent' presheaf[3]). In other words, this means that there is at most one way to glue.
(2) Given $f_i \in \mathcal{F}(U_i) \forall i$ such that $f_i|_{u_i\cap u_j} = f_j|_{u_i\cap u_j}$, there is some $f \in \mathcal{F}(U)$ such that $\mathrm{res}^u_{ui} f = f_i$.
    This axiom is called *gluibility axiom*. This means that if we have local functions and they overlap on the intersection of covers, then we can piece together the local sections to a bigger one. Also, the gluibility means there is at least one way to glue.

Remark 7.10. Sometimes, it is also taken as an ad-hoc axiom that $\mathcal{F}(\varnothing)$ is one-element set and it is the terminal object in the category where sheaf takes value. However, it is not explicitly written, for instance, in SGA 4, it is naturally embedded into the general descent/sheaf condition applied to the empty covering family on the empty set. We will see how that happens when discussing sites and topos.

Definition 7.11. A morphism of sheaves $\alpha : \mathcal{F} \to g$ is an isomorphism if there exists $\beta : \mathcal{G} \to \mathcal{F}$ such that $\alpha \circ \beta = id_{\mathcal{G}}$ and $\beta \circ \alpha = id_{\mathcal{F}}$.

Remark 7.12. A morphism of sheaves on topological space is an isomorphism if and only if it is injective and surjective.

Proof. Let $\alpha : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. For the if direction: let us assume α is an isomorphism. Then $\exists \beta : \mathcal{G} \to \mathcal{F}$ such $\alpha \cdot \beta = id_{\mathcal{G}}$ and $\beta \cdot \alpha = id_{\mathcal{F}}$. For $U \subset X$, we have $\alpha_U : \mathcal{F}(U) \to \mathcal{G}(U)$ and $\beta_U : \mathcal{G}(U) \to \mathcal{F}(U)$ which are mutually inverse and thus, α is an injective and surjective.

For the only if direction: let us assume that $\alpha : \mathcal{F} \to \mathcal{G}$ is an injective and surjective map. We can assume a morphism $\beta = \alpha^{-1} : \mathcal{G} \to \mathcal{F}$ and verify if it a morphism of sheaves. For $V \subseteq U$, we have

$$\mathrm{res}^{V,\mathcal{F}}_U : \mathcal{F}(U) \to \mathcal{F}(V), \quad \mathrm{res}^{V,\mathcal{G}}_U : \mathcal{G}(U) \to \mathcal{G}(V) \tag{198}$$

[2]A functor category $[\mathcal{C}, \mathcal{D}]$ is collection of functors $\mathcal{F} : \mathcal{C} \to \mathcal{D}$ and the morphisms are natural transformations between functors.

[3]As G. R. Kempf does in Algebraic Varieties [**33**].

or

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\alpha_U} & \mathcal{G}(U) \\ {\scriptstyle \mathrm{res}_U^{V,\mathcal{F}}}\downarrow & & \downarrow{\scriptstyle \mathrm{res}_U^{V,\mathcal{G}}} \\ \mathcal{F}(V) & \xrightarrow[\alpha_V]{} & \mathcal{G}(V) \end{array}$$

such that

$$\mathrm{res}_U^{V,\mathcal{G}} \circ \alpha_U = \alpha_V \circ \mathrm{res}_U^{V,\mathcal{F}} \tag{199}$$

and now compose with β_U and β_V on both sides

$$\beta_V \circ \mathrm{res}_U^{V,\mathcal{G}} = \mathrm{res}_U^{V,\mathcal{F}} \circ \beta_U \tag{200}$$

which is saying that β_U commutes with restriction maps and hence, define a morphism of sheaves $\beta : \mathcal{G} \to \mathcal{F}$. By construction, we have $\alpha \circ \beta = id_{\mathcal{G}}$ and $\beta \circ \alpha = id_{\mathcal{F}}$ and so α is an isomorphism. □

Definition 7.13 (Germs). Let $U, V \subset X$ and $\mathcal{F}$ be a (pre)sheaf. If we have two sections $f \in \mathcal{F}(U)$ and $g \in \mathcal{F}(V)$, and if they agree to an open neighbourhood $W \subseteq U \cap V$ through

$$f|_W = g|_W \tag{201}$$

then they are called germs.

In other words, two pairs (f, U) and (g, V) are equivalent are called germ at $p \in X$ if they satisfy (201) for an open neighborhood $W \subseteq U \cap V$ of a point p.

Definition 7.14 (Stalks). A stalk $\mathcal{F}_p$ at $p \in X$ is the collection of germs at p.

7.15. A very interesting method of studying a presheaf $\mathcal{F}$ is by studying the sections over smaller and smaller neighborhoods of a point $x \in X$.

Definition 7.16 (One more definition of Stalks). We define the stalk at point x as $\mathcal{F}_x$ to be $\varinjlim_{x \in U} \mathcal{F}(U)$ for $x \in U$ where $U's$ are the open neighborhood of x which are partially ordered by the inclusion[4].

7.17. Given a morphism of presheaves $\alpha : \mathcal{F} \to \mathcal{G}$, there is a family of morphisms of stalks $\alpha_x : \mathcal{F}_x \to g, \forall x \in X$.

Definition 7.18 (A random but important definition here of étale space). Let X be a topological space and $\mathcal{F}$ be a sheaf of sets over X. We define the *étalé space* of $\mathcal{F}$ as

$$E(\mathcal{F}) = \coprod_{x \in X} \mathcal{F}_x \tag{202}$$

[4]The philosophy is that $\mathcal{F}_x$ is the disjoint union $\bigcup \mathcal{F}(U)$ of all the sections of $\mathcal{F}$ modulo equivalent sections (germs).

with a projection $p : E(\mathcal{F}) \to X, (x, s_x) \mapsto x$ where s_x is the germ at x and the topology is given my subsets $\widetilde{s}(U) := \{(x, s_x) : x \in U\}$ where $U \subset X$ are open and s is a section in $\mathcal{F}(U)$.

The projection is a local homeomorphism.

7.19. We can construct the sheaf from the continuous sections of $p : E(\mathcal{F}) \to X$ over open subsets of X.

The point that the sheaf as a topological space assembled from its stalks is given by above definition. However, as mentioned in EGA 3.1.6, one does not work with sheaf as a topological space when (the sheaf on a category) $\mathcal{C}$ has additional structures on morphism.

Definition 7.20 (Kernels). For the morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$, we define the presheaf kernel of of presheaves is

$$(\mathrm{Ker}_{pre}\phi)(U) = \mathrm{Ker}\phi(U) \tag{203}$$

and in the the respect of restriction maps

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Ker}_{pre}\phi(U) & \longrightarrow & \mathcal{F}(U) & \xrightarrow{\alpha(U)} & \mathcal{G}(U) \\ & & \vdots & & \downarrow{\scriptstyle res_U^V} & & \downarrow{\scriptstyle res_U^V} \\ 0 & \longrightarrow & \mathrm{Ker}_{pre}\phi(V) & \longrightarrow & \mathcal{F}(V) & \xrightarrow[\alpha(V)]{} & \mathcal{G}(V) \end{array}$$

Lemma 7.21. If $\mathcal{F}$ and $\mathcal{G}$ are sheaves, then $\mathrm{Ker}_{pre}\phi$ is a sheaf $\mathrm{Ker}\phi$.

Proof. Since $\mathcal{F}$ is a sheaf, $f_1, f_2 \in \mathrm{Ker}_{pre}\phi(U)$ agree on an open cover.

Now assume $f_i \in Ker_{pre}\phi(U_i)$ and they agree on overlaps since $\mathcal{F}$ is a sheaf. We glue together to get a section $f \in \mathcal{F}(U)$.

$$\phi(U_i)(f_i) = \phi(U)(f)|_{U_i} \tag{204}$$

and U_i are in the kernel. Therefore, $\phi(U)(f) = 0$ by gluing.

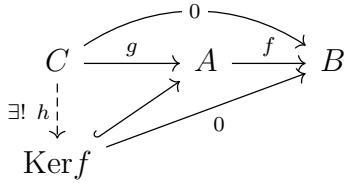
In simple words, since $\mathrm{Ker}_{pre}\phi$ is subsheaf[5] of sheaf $\mathcal{F}$, the sheaf structure is automatically inherited. □

Remark 7.22. In a category $\mathcal{C}$ with a zero object[6], we know by the universal property that for morphism $f : A \to B$, there exists a morphism $g : C \to A$ such that $f \circ g = 0$ is

[5]A subsheaf is an injective sheaf morphism $\mathcal{F} \hookrightarrow \mathcal{G}$.

[6]This implies that we have zero morphisms.

zero morphism and there is a unique morphism $C \to \mathrm{Ker}(f)$ such that



(More about this will be discussed in Chapter. 4.) But with that remark, one can note that the sheaf Kerϕ also follows this universal property in *category of presheaves*.

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